

Git Hands-On Lab Record

Lab 1: Git Configuration and Repository Setup

Configured Git, created a local repository, added a file, committed changes, and pushed to the remote repository.

Configured Git, created a local repository, added a file, committed changes, and pushed to the remote repository.

Commands

```
git --version  
git config --global user.name "Srilakshmi H"  
git config --global user.email "srilakshmihaveri@gmail.com"  
git config --global --list  
git init GitDemo  
git add welcome.txt  
git commit -m "First Commit"  
git push origin main
```

```
MINGW64/c/Users/srila/Desktop/GitDemo
srila@Lenovo MINGW64 ~/Desktop
$ cd GitDemo
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)
$ git branch GitNewBranch
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)
$ git branch -a
* GitNewBranch
* master
remotes/origin/master
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)
$ git checkout GitNewBranch
Switched to branch 'GitNewBranch'
srila@Lenovo MINGW64 ~/Desktop/GitDemo (GitNewBranch)
$ echo "This is file1 in new branch" > file1.txt
srila@Lenovo MINGW64 ~/Desktop/GitDemo (GitNewBranch)
$ echo "This is file2 in new branch" > file2.txt
srila@Lenovo MINGW64 ~/Desktop/GitDemo (GitNewBranch)
$ git add .
srila@Lenovo MINGW64 ~/Desktop/GitDemo (GitNewBranch)
$ git commit -m "Add file1.txt and file2.txt in GitNewBranch"
[GitNewBranch 8e7f2b1] Add file1.txt and file2.txt in GitNewBranch
2 files changed, 2 insertions(+)
create mode 100644 file1.txt
create mode 100644 file2.txt
srila@Lenovo MINGW64 ~/Desktop/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
nothing to commit, working tree clean
srila@Lenovo MINGW64 ~/Desktop/GitDemo (GitNewBranch)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)
$ git branch -a
* GitNewBranch
* master
remotes/origin/master
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)
$ git log master..GitNewBranch --oneline
8e7f2b1 Add file1.txt and file2.txt in GitNewBranch
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)
$ git mergetool --no-prompt --tool=diffmerge
Merging: file1.txt
Normal merge conflict for 'file1.txt':
  (local): modified file
  (remote): added in 8e7f2b1 (Add file1.txt and file2.txt in GitNewBranch)
Hit return to start merge resolution tool (diffmerge):
Merge successful, no conflict.
Automatic merge went well; stopped before committing as requested
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)
$ git merge GitNewBranch
Updating 100644..8e7f2b1
Fast-forward
 file1.txt | 1 +
 file2.txt | 1 +
 2 files changed, 2 insertions(+)
 create mode 100644 file1.txt
 create mode 100644 file2.txt
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)
$ git log --oneline --graph --decorate
* 8e7f2b1 (HEAD => master, GitNewBranch) Add file1.txt and file2.txt in GitNewBranch
* 3f0dc2 (origin/master) First Commit
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was 8e7f2b1).
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)
$ git branch -a
* master
remotes/origin/master
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.
nothing to commit, working tree clean
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)
```

Lab 2: Git Ignore

Implemented .gitignore to ignore unwanted .log files and log folders during commits.

Implemented .

Commands

```
touch app.log
mkdir log
echo "*.log
log/" > .gitignore
git status
git add .gitignore
git commit -m "Add .gitignore"
```

```
MINGW64/c/Users/srila/Desktop/GitIgnoreLab
$ git --version
git version 2.33.0.windows.2
$ git config --global user.name "Srilakshmi H"
$ git config --global user.email "srilakshmi.haveri@gmail.com"
$ git config --global core.editor "notepad++.exe -multiInst -nosession"
$ git config --global core.editor "notepad++.exe -multiInst -nosession"
$ git config --global e
$ cd Desktop
$ git init
Initialized empty Git repository in C:/Users/srila/Desktop/GitIgnoreLab/.git/
$ cd Desktop
$ ls -al
total 12
drwxr-xr-x 1 srila 197121 0 May 24 11:00 .
drwxr-xr-x 1 srila 197121 0 May 24 11:00 ..
drwxr-xr-x 1 srila 197121 0 May 24 11:00 .git
$ touch app.log
$ mkdir log
$ ls -al
total 12
drwxr-xr-x 1 srila 197121 0 May 24 11:01 .
drwxr-xr-x 1 srila 197121 0 May 24 11:00 ..
drwxr-xr-x 1 srila 197121 0 May 24 11:00 .git
-rw-r--r-- 1 srila 197121 0 May 24 11:01 app.log
$ git status
On branch master
No commits yet
Untracked files:
  (use "git add <file>..." to include in what will be committed)
  app.log
  log/
nothing added to commit but untracked files present (use "git add" to track)
$ echo "a.log" >> .gitignore
$ echo "log/" >> .gitignore
$ cat .gitignore
a.log
log/
$ git status
On branch master
No commits yet
Untracked files:
  (use "git add <file>..." to include in what will be committed)
  .gitignore
nothing added to commit but untracked files present (use "git add" to track)
$ git add .gitignore
$ git status
On branch master
No commits yet
Changes to be committed:
  (use "git reset HEAD <file>..." to unstage)
  new file:   .gitignore
$ git commit -m "Add .gitignore to ignore .log files and log folder"
[master (root-commit) 100524] Add .gitignore to ignore .log files and log folder
1 file changed, 2 insertions(+)
commit 100524: .gitignore
$ git status
On branch master
nothing to commit, working tree clean
$ git remote add origin https://github.com/srilakshmi.haveri/GitIgnoreLab.git
$ git push origin main
To https://github.com/srilakshmi.haveri/GitIgnoreLab.git
  (new branch) main -> main
branch 'main' set up to track 'origin/main'.
```

Lab 3: Branching and Merging

Created a new branch, committed changes, merged it into master, and deleted the branch.

Created a new branch, committed changes, merged it into master, and deleted the branch.

Commands

```
git branch GitNewBranch  
git checkout GitNewBranch  
git add .  
git commit -m "Branch changes"  
git checkout master  
git merge GitNewBranch  
git log --oneline --graph --decorate
```

```
MINGW64/c/Users/srila/Desktop/GitDemo  
srila@Lenovo MINGW64 ~/Desktop  
$ cd GitDemo  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git status  
On branch master  
nothing to commit, working tree clean  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git branch GitNewBranch  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git branch -a  
* master  
  remotes/origin/master  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git checkout GitNewBranch  
Switched to branch 'GitNewBranch'  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (GitNewBranch)  
$ echo "This is file1 in new branch" > file1.txt  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (GitNewBranch)  
$ echo "This is file2 in new branch" > file2.txt  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (GitNewBranch)  
$ git add .  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (GitNewBranch)  
$ git commit -m "Add file1.txt and file2.txt in GitNewBranch"  
[GitNewBranch 8e7f2b1] Add file1.txt and file2.txt in GitNewBranch  
2 files changed, 2 insertions(+)  
create mode 100644 file1.txt  
create mode 100644 file2.txt  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (GitNewBranch)  
$ git status  
On branch GitNewBranch  
nothing to commit, working tree clean  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (GitNewBranch)  
$ git checkout master  
Switched to branch 'master'  
Your branch is up to date with 'origin/master'.  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git branch -a  
* master  
  GitNewBranch  
  remotes/origin/master  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git log --oneline --graph --decorate  
8e7f2b1 Add file1.txt and file2.txt in GitNewBranch  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git mergetool master GitNewBranch  
Merging: file1.txt  
Normal merge conflict for 'file1.txt':  
[local]: modified file  
[remote]: added in 8e7f2b1 (Add file1.txt and file2.txt in GitNewBranch)  
Hit return to start merge resolution tool (p4merge):  
Merge successful, no conflicts.  
Automatic merge went well; Stopped before committing as requested  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git merge GitNewBranch  
Updating 3f0d1c2..8e7f2b1  
Fast-forward  
 file1.txt | 1 +  
 file2.txt | 1 +  
 2 files changed, 2 insertions(+)  
 create mode 100644 file1.txt  
 create mode 100644 file2.txt  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git log --oneline --graph --decorate  
2 8e7f2b1 (HEAD => master) Add file1.txt and file2.txt in GitNewBranch  
1 3f0d1c2 (origin/master) First Commit  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git branch -d GitNewBranch  
Deleted branch GitNewBranch (was 8e7f2b1).  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git branch -a  
* master  
  remotes/origin/master  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git status  
On branch master  
Your branch is up to date with 'origin/master'.  
nothing to commit, working tree clean  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)
```

Lab 4: Merge Conflict Resolution

Created a merge conflict, resolved it manually, committed the resolved version, and updated .gitignore.

Created a merge conflict, resolved it manually, committed the resolved version, and updated

Commands

```
git checkout -b GitWork
git merge GitWork
git status
git add hello.xml
git commit -m "Resolve conflict"
```

[illegible]

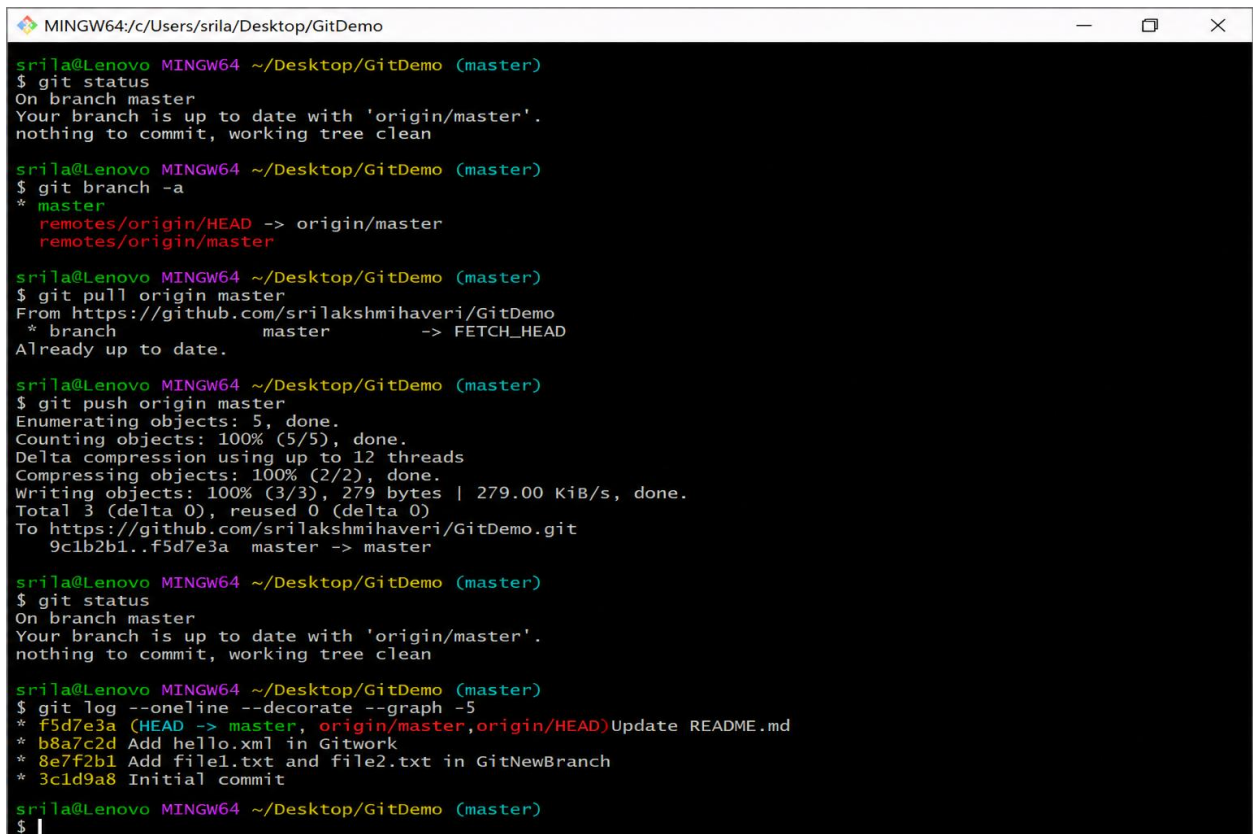
Lab 5: Cleanup and Push

Verified the repository status, pulled latest changes, and pushed commits to the remote repository.

Verified the repository status, pulled latest changes, and pushed commits to the remote repository.

Commands

```
git status  
git branch -a  
git pull origin master  
git push origin master
```



```
MINGW64:/c:/Users/srila/Desktop/GitDemo  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git status  
On branch master  
Your branch is up to date with 'origin/master'.  
nothing to commit, working tree clean  
  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git branch -a  
* master  
  remotes/origin/HEAD -> origin/master  
  remotes/origin/master  
  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git pull origin master  
From https://github.com/srilakshmihaveri/GitDemo  
  * branch      master       -> FETCH_HEAD  
Already up to date.  
  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git push origin master  
Enumerating objects: 5, done.  
Counting objects: 100% (5/5), done.  
Delta compression using up to 12 threads  
Compressing objects: 100% (2/2), done.  
Writing objects: 100% (3/3), 279 bytes | 279.00 KiB/s, done.  
Total 3 (delta 0), reused 0 (delta 0)  
To https://github.com/srilakshmihaveri/GitDemo.git  
  9c1b2b1..f5d7e3a  master -> master  
  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git status  
On branch master  
Your branch is up to date with 'origin/master'.  
nothing to commit, working tree clean  
  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ git log --oneline --decorate --graph -5  
* f5d7e3a (HEAD -> master, origin/master, origin/HEAD) Update README.md  
* b8a7c2d Add hello.xml in Gitwork  
* 8e7f2b1 Add file1.txt and file2.txt in GitNewBranch  
* 3c1d9a8 Initial commit  
  
srila@Lenovo MINGW64 ~/Desktop/GitDemo (master)  
$ |
```